

VIGNESH GOPIKRISHNAN

Full-Stack Engineer · AI Systems · Backend Infrastructure

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PROFESSIONAL SUMMARY

Full-stack engineer (MCA, RVITM Bangalore) with production experience in Node.js, Python, TypeScript, and React. Built AI-integrated applications, async pipelines, and scalable REST APIs. Active open-source contributor working on AI agent infrastructure. Passionate about building resilient backend systems for AI-powered products at enterprise scale.

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, C/C++

Backend & APIs: Node.js, Express.js, REST APIs, WebSockets

Databases: PostgreSQL, MongoDB, MySQL, SQLite, Redis (caching patterns)

AI / Agent Systems: MCP (Model Context Protocol), LLM integration, RAG-aware tooling

Cloud & DevOps: AWS, Vercel, Docker, Kubernetes, GitHub Actions CI/CD

EXPERIENCE

Open-Source Contributor | Multiple Organizations | Remote

2025 – Present

- Contributing to open-source projects across multiple organizations, including AI agent infrastructure and developer tooling, collaborating with distributed teams on production-grade codebases.
- Working within codebases enforcing strict typing, linting, and pre-commit hooks; gaining hands-on experience with AI agent orchestration patterns, async pipelines, and observability integrations.

Backend Engineer Intern | Mantarlee Social Innovations Federation | Bangalore

Oct 2024 – Nov 2024

- Engineered backend services using Node.js, Express.js, and MySQL; achieved a 15% improvement in server response time for the company's prototype application.
- Architected and optimised the MySQL database schema, resulting in a 20% increase in data retrieval efficiency; directly improving application scalability and performance.
- Developed and refined RESTful API endpoints and collaborated with a 3-member team to deliver seamless API-to-frontend integration with full data consistency.

PERSONAL PROJECTS

Stackforge — AI Agent Orchestration System

[GitHub](#) · [Live](#) | Jan 2026 – Mar 2026

- Engineered a multi-agent orchestration system in TypeScript/Express.js that scaffolds full-stack project blueprints from plain-English prompts via 6 coordinated sub-agents.
- Applied per-agent token optimisation including input compression, output caps, and budget guardrails, demonstrating async pipeline design at scale.
- Built a monorepo with strict TypeScript and automated typecheck/test pipelines to enforce production-grade code quality across all packages.
- Deployed on Vercel with CI/CD linked to GitHub; managed 2 tracked production deployments with zero-downtime release patterns.

Fossel — Local-First MCP Memory Server

[GitHub](#) · [Live](#) | Jan 2026 – Present

- Built a TypeScript/SQLite MCP memory server for open-source contributors, enabling persistent full-text search of repo-specific context; published to npm as fossel (v1.0.6) with 4 tool endpoints exposed via a stdio interface compatible with AI coding tools like Cursor and Claude Desktop.
- Designed an efficient data storage layer for fast indexed retrieval, applying scalable, maintainable data storage principles for AI agent memory and context systems.
- Automated builds and releases with GitHub Actions CI/CD; maintained a clean ESM TypeScript codebase bundled with tsup and deployed the companion site on Vercel.

Folio — Browser-Based PDF Workspace

[GitHub](#) · [Live](#) | Jan 2026 – Mar 2026

- Built a fully client-side PWA in React, TypeScript, and Vite supporting page viewing, annotation, form editing, merge/split, and full-text search; zero backend footprint.
- Integrated pdf.js and Fabric.js for rich PDF rendering and canvas-based overlay annotations; configured Workbox service workers for offline-ready performance.

Skannr — Universal AI Code Analyzer

[GitHub](#) · [Live](#) | 2026

- Built a TypeScript-based universal code analysis engine that scans large codebases and generates compressed structural "skeletons," cutting token usage by ~96.5% versus full-file scans for AI agent context.
- Designed a hybrid file-ranking system combining lexical matching, import/export graph structure, and dependency-graph centrality with a cross-rerank pass, achieving ~75% top-1 keyword coverage on real-world queries.
- Shipped both a CLI and an MCP (Model Context Protocol) server integration for Gemini CLI, exposing a scan_codebase tool with grounded, evidence-cited responses and a hybrid in-memory/disk cache layer.

EDUCATION

Master of Computer Applications (MCA) | RV Institute of Technology and Management | Bangalore

2024 – 2026 · CGPA: 8.6 · Karnataka

Bachelor of Computer Applications (BCA) | St Joseph's University | Bangalore

2021 – 2024 · CGPA: 7.8 · Karnataka

CERTIFICATIONS

- AWS Certified Cloud Practitioner
- Full Stack Web Development Bootcamp, Udemy (20+ projects built)
- Dynamics and Interactive Website Using React, SJU
- Problem Solving (Basics), HackerRank